AD23431 - STATISTICAL ANALYSIS AND COMPUTING

EXP 8: Statistical Testing Using R

T-Test: Identify the Appropriate Test and Implement in R

1. ANOVA (One-Way ANOVA)

A college wants to check whether the average marks differ across three departments (CSE, ECE, MECH). A sample of students' marks is collected.

- Identify the test
- Null Hypothesis (H_0): Mean marks of all departments are equal
- Alternate Hypothesis (H_1): At least one department mean differs

PROGRAM:

```
data <- read.csv("C:\\Users\\TARRUN\\Downloads\\data.csv")
result <- aov(Marks ~ Department, data = data)
summary(result)
```

OUTPUT:

```
            Df Sum Sq Mean Sq F value  Pr(>F)
Department   2  843.6   421.8    13.7 0.00579 **
Residuals    6  184.7    30.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2. F-Test (Variance Comparison)

A researcher wants to compare whether variance in scores of two classes is equal.

- Identify the test
- H_0: Variances are equal
- H_1: Variances are not equal

PROGRAM:

```
data <- read.csv("C:\\Users\\TARRUN \\Downloads\\data.csv")
A <- subset(data, Class=="A")$Marks
B <- subset(data, Class=="B")$Marks
print(var.test(A, B))
```

OUTPUT:

```
        F test to compare two variances

data:  A and B
F = 2.7949, num df = 2, denom df = 2, p-value = 0.527
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
  0.07166338 109.00000000
sample estimates:
ratio of variances
        2.794872
```

3. Mann–Whitney U Test (Wilcoxon Rank-Sum Test)

A teacher compares scores of students taught using two different methods, but data is not normally distributed.

- Identify the test
- H_0: Both groups have the same distribution
- H_1: Distributions differ

PROGRAM:

```
data <- read.csv("C:\\Users\\TARRUN\\Downloads\\data.csv")
m1 <- subset(data, Method==1)$Marks
m2 <- subset(data, Method==2)$Marks
wilcox.test(m1, m2)
```

OUTPUT:

```
        Wilcoxon rank sum exact test

data:  m1 and m2
W = 9, p-value = 0.1
alternative hypothesis: true location shift is not equal to 0
```

4. Fisher's Exact Test

A study checks whether pass/fail outcome depends on gender.

	Pass	Fail
Male	8	2
Female	4	6

- Identify the test: Fisher's Exact Test
- H_0: No association between gender and result
- H_1: Association exists

PROGRAM:

```
data <- read.csv("C:\\Users\\TARRUN\\Downloads\\data.csv")
table_data <- table(data$Gender, data$Result)
print(table_data)
result <- fisher.test(table_data)
print(result)
```

OUTPUT:

```
       Fail Pass
  Female    6    4
  Male      2    8

        Fisher's Exact Test for Count Data

data:  table_data
p-value = 0.1698
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
  0.6026805 79.8309210
sample estimates:
odds ratio
  5.430473
```

5. Kruskal–Wallis Test

A researcher compares student performance across three teaching methods, but data is not normal.

- Identify the test: Kruskal–Wallis Test
- H_0: All groups have same distribution
- H_1: At least one group differs

PROGRAM:

```
data <- read.csv("C:\\Users\\TARRUN\\Downloads\\data.csv")
print(kruskal.test(Marks ~ Method, data = data))
```

OUTPUT:

Kruskal-Wallis rank sum test

data: Marks by Method
Kruskal-Wallis chi-squared = 6.4889, df = 2, p-value = 0.03899

6. Bartlett's Test

A researcher checks whether variances across multiple groups are equal before applying ANOVA.

- Identify the test: Bartlett's Test
- H_0: Variances are equal across groups
- H_1: Variances differ

PROGRAM:

```
data <- read.csv("C:\\Users\\TARRUN\\Downloads\\data.csv")
print(bartlett.test(Marks ~ Group, data = data))
```

OUTPUT:

Bartlett test of homogeneity of variances

data: Marks by Group
Bartlett's K-squared = 0.59154, df = 2, p-value = 0.744

7. Binomial Distribution

A quiz consists of 10 multiple-choice questions. Each question has 4 options, and only one is correct. A student guesses randomly.

1. Find the probability that the student gets exactly 4 correct answers.
2. Find the probability of getting at most 3 correct answers.
3. Simulate 1000 such students and plot the distribution.

PROGRAM:

```
dbinom(4, size=10, prob=0.25)
pbinom(3, size=10, prob=0.25)
x <- rbinom(1000, size=10, prob=0.25)
hist(x)
```

OUTPUT:

0.145

0.776

8. Poisson Distribution

A call center receives an average of 5 calls per hour.

1. Find the probability of receiving exactly 3 calls in an hour.
2. Find the probability of receiving at most 4 calls.
3. Generate and plot 1000 observations.

PROGRAM:

```
dpois(3, lambda=5)
ppois(4, lambda=5)
x <- rpois(1000, lambda=5)
hist(x)
```

OUTPUT:

0.140
0.440

9. Normal Distribution

The marks of students in an exam follow a normal distribution with mean = 70 and standard deviation = 10.

1. Find the probability that a student scores less than 60.
2. Find the probability that a student scores between 60 and 80.
3. Generate 1000 values and plot the distribution.

PROGRAM:

```
pnorm(60, mean=70, sd=10)
pnorm(80,70,10) - pnorm(60,70,10)
x <- rnorm(1000, mean=70, sd=10)
hist(x)
```

OUTPUT:

0.1587
0.6926